COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 009B-CPE21S4	**SECTION:**	CPE21S4
DATE PERFORMED:	4/09/2026	**DATE SUBMITTED:**	4/09/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

main.cpp | sortingAlgo.h

7.1 > main.cpp > main()

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
#include "sortingAlgo.h"

const int maxSize = 100;

void displayArr(int arr[]);

int main(){

    //create an array with unsorted list with 100 maximum size
    int unsortedArr[maxSize];

    //use time (0) as seed for random values
    srand(time(0));

    //populate the array with random values
    for(int i = 0; i < maxSize; i++){
        unsortedArr[i] = rand() %5 +1;
    }

    displayArr(unsortedArr);

    //std::cout<<"\n\n Bubble Sort testing: \n";
    //bubbleSort(unsortedArr, maxSize);
    //displayArr(unsortedArr);

    std::cout<<"\n\n selectionSort testing: \n";
    selectionSort(unsortedArr, maxSize);
    displayArr(unsortedArr);

    // std::cout<<"\n\n insertionSort testing: \n";
    // insertionSort2(unsortedArr, maxSize);
    // displayArr(unsortedArr);


}

void displayArr(int arr[]){
    for (int i = 0; i < maxSize; i++){
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
}
```

main.cpp × sortingAlgo.h ×

7.1 > sortingAlgo.h > bubbleSort<T>(T [], int)

```cpp
#ifndef SORTINGALGO_H
#define SORTINGALGO_H

#include <iostream>
using namespace std;

template<typename T>
void bubbleSort(T arr[], int maxSize){
    for(int i = 0; i < maxSize; i++){
        for(int j = i + 1; j < maxSize; j++){
            if(arr[i] > arr[j]){
                std::swap(arr[j], arr[i]);
            }
        }
    }
}

template<typename T>
int routineSmallest(T arr[], int k, int maxSize){
    int position;
    int smallestElem = arr[k];
    position = k;

    for(int j = k + 1; j < maxSize; j++){
        if(arr[j] < smallestElem){
            smallestElem = arr[j];
            position = j;
        }
    }
    return position;
}

template<typename T>
void selectionSort(T arr[], int maxSize){
    int POS;
    for(int i = 0; i < maxSize; i++){
        POS = routineSmallest(arr, i, maxSize);
        std::swap(arr[i], arr[POS]);
    }
}

template<typename T>
void insertionSort2(T arr[], int maxSize){
    int k, j, temp;
    while (k < maxSize){
        temp = arr[k];

        // The rest follows your structure — complete version below
        for (j = k - 1; j >= 0 && temp < arr[j]; j--){
            arr[j + 1] = arr[j];
        }
        arr[j + 1] = temp;
        k++;
    }
}
```

```
[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\" && g++ main.cpp -o main && "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\"main
2 2 1 2 4 1 4 4 5 3 3 4 4 5 3 3 2 4 1 5 5 2 2 1 4 1 2 1 4 4 5 5 4 5 3 3 1 4 5 3 4 2 5 3 1 4 3 2 1 5 4 2 1 2 2 4 5 4 5 4 5 3 3 5 4 4 4 4 1 2 1 4 3 3 3 2 2 2 4 4 1 1 4 2 4 5 2 1 2 1 4 1 1 1 5 3 1 2 1 5

 Bubble Sort testing:
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

[Done] exited with code=0 in 1.281 seconds
```

```
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2 1 3 1 5 4 3 5 3 3 5 4 2 5 4 1 5 3 4 1 4 2 1 2 2 4 3 5 2 1 1 1 5 2 1 5 4 1 1 1 4 2 5 2 1 5 4 5 3 5 2 5 2 5 2 2 4 1 3 5 5 5 2 4 1 2 2 3 5 3 4 5 4 1 1 2 2 5 5 1 5 2 4 2 3 5 4 5 3 2 2 1 3 3 4 2 4 4 1 1

 selectionSort testing:
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

[Done] exited with code=0 in 1.104 seconds
```

```
[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\" && g++ main.cpp -o main && "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\"main
4 5 1 1 4 2 3 5 5 4 2 1 1 5 3 3 4 3 2 5 3 2 4 2 3 4 5 2 3 5 2 2 5 1 3 3 1 1 4 2 2 4 5 1 4 5 1 4 1 4 3 5 2 1 1 2 4 2 1 4 2 1 1 4 3 4 1 2 1 4 3 2 4 4 1 4 1 1 5 5 4 3 5 5 1 5 1 2 2 2 1 4 2 5 2 1 1 1 2 1

 insertionSort testing:
4 5 1 1 4 2 3 5 5 4 2 1 1 5 3 3 4 3 2 5 3 2 4 2 3 4 5 2 3 5 2 2 5 1 3 3 1 1 4 2 2 4 5 1 4 5 1 4 1 4 3 5 2 1 1 2 4 2 1 4 2 1 1 4 3 4 1 2 1 4 3 2 4 4 1 4 1 1 5 5 4 3 5 5 1 5 1 2 2 2 1 4 2 5 2 1 1 1 2 1

[Done] exited with code=0 in 1.061 seconds
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

main.cpp × sortingAlgo.h ×

7.1 > supp > main.cpp > selectionsort(int [], int)

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
#include "sortingAlgo.h"

using namespace std;

const int maxSize = 100;

void selectionsort(int values[], int size)
{
    for (int i = 0; i < size - 1; ++i)
    {
        int smallest = i;
        for (int j = i + 1; j < size; ++j)
        {
            if (values[j] < values[smallest])
            {
                smallest = j;
            }
        }

        int temporary = values[i];
        values[i] = values[smallest];
        values[smallest] = temporary;
    }
}

string candinates[6] = {
    "",
    "Bo Dalton Capistrano",
    "Cornelius Raymon Agustín",
    "Deja Jayla Bañaga",
    "Lalla Brielle Yabut",
    "Franklin Relano Castro"};

int main()
{
    int votes[maxSize];
    int voteCount[6] = {0};

    srand(time(0));
    cout << " VOTES (Unsorted)\n";
    for (int i = 0; i < maxSize; i++)
    {
        votes[i] = rand() % 5 + 1;
```

```
48      }
49      cout << "\n\n";
50
51      selectionsort(votes, maxSize);
52      cout << " VOTES (Sorted)\n";
53      for (int i = 0; i < maxSize; i++)
54      {
55          cout << votes[i] << " ";
56      }
57      cout << "\n\n";
58
59      for (int i = 0; i < maxSize; i++)
60      {
61          voteCount[votes[i]]++;
62      }
63
64  //votes count
65      cout << " VOTES COUNT\n";
66      int highestVoteCount = 0;
67      int winnerIndex = 0;
68
69      for (int i = 1; i <= 5; i++) {
70          cout << "Candidate " << i << ": " << candinates[i] << endl;
71          cout << "   Votes: " << voteCount[i] << " / " << maxSize << endl;
72
73          if (voteCount[i] > highestVoteCount) {
74              highestVoteCount = voteCount[i];
75              winnerIndex = i;
76          }
77          cout    << endl;
78      }
79
80      cout << "Winner: " << candinates[winnerIndex] << endl;
81      cout << "Total Votes: " << highestVoteCount << endl;
82
83      return 0;
84
85  }
```

main.cpp sortingAlgo.h

7.1 > supp > sortingAlgo.h > ...

```cpp
#ifndef SORTINGALGO_H
#define SORTINGALGO_H

#include <iostream>
using namespace std;

template <typename T>
int routineSmallest(T arr[], int start, int maxSize) {
    int pos = start;
    for (int j = start + 1; j < maxSize; j++) {
        if (arr[j] < arr[pos])
            pos = j;
    }
    return pos;
}

template <typename T>
void selectionSort(T arr[], int maxSize) {
    for (int i = 0; i < maxSize; i++) {
        int POS = routineSmallest(arr, i, maxSize);
        swap(arr[i], arr[POS]);
    }
}

#endif
```

```
[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\supp\" && g++ main.cpp -o main && "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midterms\7.1\supp\"main
 VOTES (Unsorted)
3 4 4 2 3 1 5 2 4 5 4 5 3 4 5 3 3 1 5 5 3 1 3 5 4 3 4 1 3 2 4 5 3 3 4 4 2 4 5 3 2 2 1 5 2 5 4 3 4 5 3 1 5 4 5 3 2 5 2 3 5 2 3 1 1 4 1 1 2 1 2 5 3 1 3 1 3 5 3 1 5 3 3 2 2 3 2 1 1 4 5 5 4 5 4 2 2 1 5 3

 VOTES (Sorted)
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

 VOTES COUNT
Candidate 1: Bo Dalton Capistrano
   Votes: 17 / 100

Candidate 2: Cornelius Raymon Agustin
   Votes: 17 / 100

Candidate 3: Deja Jayla Bañaga
   Votes: 25 / 100

Candidate 4: Lalla Brielle Yabut
   Votes: 18 / 100

Candidate 5: Franklin Relano Castro
   Votes: 23 / 100

Winner: Deja Jayla Bañaga
Total Votes: 25

[Done] exited with code=0 in 0.734 seconds
```

Manual count
Candidate 1: Bo Dalton Capistrano

Votes: 17 / 100

Candidate 2: Cornelius Raymon Agustín
Votes: 17 / 100

Candidate 3: Deja Jayla Bañaga
Votes: 25 / 100

Candidate 4: Lalla Brielle Yabut
Votes: 18 / 100

Candidate 5: Franklin Relano Castro
Votes: 23 / 100

Count Result of Algorithm
All rhe same as output console showing sorted array and manual count

Question: Was your developed vote counting algorithm effective? Why or why not?
Yes, it was effective. Using selection sort from activity 7 pdf, the votes were grouped together and easy to count. The program gave correct results and showed the winner properly.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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